

Non-unimodular $\hat{1}$ SUGRA Bases: target = (-1)

$\hat{1}$ external: $C^2 = -1$, $b_0 \cdot C = -1$.
Total: 24 bases (from 24 distinct LST bases, $T_{\text{LST}} \in [0, 20]$).
Exact det = 0: 24, fractional T_{crit} : 0. Filtered: $\Delta_{\text{crit}} \leq 0$.
 $\hat{1}$ = attached external.

T	Count		T_{crit}	Count
2	1		10	17
3	2		11	3
4	2		12	2
5	2		13	2
6	3			
7	3			
8	3			
9	4			
10	4			

$ \det $	Count
4	24

$T = 2$ (1 bases)

Base: 11

$$\begin{aligned} 11 + \hat{1} \quad \Delta = -242, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 2, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 2, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}_{b_0} \end{aligned}$$

$T = 3$ (2 bases)

Base: 121

$$\begin{aligned} 121 + \hat{1} \quad \Delta = -213, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 3, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 3, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \end{aligned}$$

Base: 212

$$\begin{aligned} \hat{1} 212 \quad \Delta = -213, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 3, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 3, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \end{aligned}$$

$T = 4$ (2 bases)

Base: 1221

$$\begin{aligned} 1221 + \hat{1} \quad \Delta = -184, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 4, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \end{aligned}$$

Base: $\overset{2}{122}$

$$\begin{aligned} \overset{2}{221} \hat{1} \quad \Delta = -184, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 4, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 4, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \end{aligned}$$

$T = 5$ (2 bases)

Base: 12221

$$\begin{aligned} 12221 + \hat{1} \quad \Delta = -155, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 5, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

Base: $\overset{2}{1222}$

$$\begin{aligned} \overset{2}{2221} \hat{1} \quad \Delta = -155, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 5, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 5, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

$T = 6$ (3 bases)

Base: 122221

$$\begin{aligned} 122221 + \hat{1} \quad \Delta = -126, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 6, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0} \end{aligned}$$

Base: $\overset{2}{12222}$

$$\begin{aligned}
& \overset{2}{22221}\hat{1} \quad \Delta = -126, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0) \\
& T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1) \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

Base: $\overset{31}{1232}$

$$\begin{aligned}
& \overset{31}{2321}\hat{1} \quad \Delta = -145, |\det| = 4, (n_+, n_-, n_0) = (1, 6, 0) \\
& T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 6, 1) \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$T = 7$ **(3 bases)**

Base: 1222221

$$\begin{aligned}
& 1222221 + \hat{1} \quad \Delta = -97, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0) \\
& T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1) \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

Base: $\overset{2}{122222}$

$$\begin{aligned}
& \overset{2}{222221}\hat{1} \quad \Delta = -97, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0) \\
& T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1) \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

Base: $\overset{2}{231321}$

$$\begin{aligned}
& \overset{2}{231321}\hat{1} \quad \Delta = -117, |\det| = 4, (n_+, n_-, n_0) = (1, 7, 0) \\
& T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 7, 1) \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$T = 8$ **(3 bases)**

Base: 12222221

$$\begin{aligned}
& 12222221 + \hat{1} \quad \Delta = -68, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0) \\
& T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1) \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 6 \\ 2 & 6 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

Base: $\overset{2}{1222222}$

$$\begin{aligned}
& \overset{2}{2222221}\hat{1} \quad \Delta = -68, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0) \\
& T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1) \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 6 \\ 2 & 6 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

Base: $\overset{2}{2313221}$

$$\begin{aligned}
& \overset{2}{2313221}\hat{1} \quad \Delta = -88, |\det| = 4, (n_+, n_-, n_0) = (1, 8, 0) \\
& T_{\text{crit}} = 11, (n_+, n_-, n_0)_{\text{crit}} = (1, 8, 1) \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

$T = 9$ **(4 bases)**

Base: 122222221

$$\begin{aligned}
& 122222221 + \hat{1} \quad \Delta = -39, |\det| = 4, (n_+, n_-, n_0) = (1, 9, 0) \\
& T_{\text{crit}} = 10, (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1) \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 7 \\ 2 & 7 & 7 \end{pmatrix}_{b_0}
\end{aligned}$$

Base: 12222222^2

$$22222221\hat{1} \quad \Delta = -39, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 9, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 7 \\ 2 & 7 & 7 \end{pmatrix}_{b_0}$$

Base: 12314132^2

$$2\hat{1}^{123141}32 \quad \Delta = -87, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 9, 0) \\ T_{\text{crit}} = 12, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

Base: $31\hat{1}^{1231}513$

$$31\hat{1}^{1231}513 \quad \Delta = -116, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 9, 0) \\ T_{\text{crit}} = 13, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 9, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$T = 10$ **(4 bases)**

Base: 122222221

$$122222221 + \hat{1} \quad \Delta = -10, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 10, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 8 \\ 2 & 8 & 8 \end{pmatrix}_{b_0}$$

Base: 12222222^2

$$22222221\hat{1} \quad \Delta = -10, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 10, 0) \\ T_{\text{crit}} = 10, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 8 \\ 2 & 8 & 8 \end{pmatrix}_{b_0}$$

Base: 122314132^2

$$2\hat{1}^{1223141}32 \quad \Delta = -58, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 10, 0) \\ T_{\text{crit}} = 12, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 6 \\ 2 & 6 & 6 \end{pmatrix}_{b_0}$$

Base: $31\hat{1}^{12231}513$

$$31\hat{1}^{12231}513 \quad \Delta = -87, \quad |\det| = 4, \quad (n_+, n_-, n_0) = (1, 10, 0) \\ T_{\text{crit}} = 13, \quad (n_+, n_-, n_0)_{\text{crit}} = (1, 10, 1) \\ \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$